undoubtedly more complicated than in Everett, and Hilbert space, the set of all possible wave functions, is as big as ever. The possibility of many worlds is not avoided by erasing the worlds (as in GRW), but simply by denying that they're real. The way Bohmian dynamics works is far from elegant. Long after classical mechanics was superseded, physicists still intuitively cling to something like Newton's third law: if one thing pushes on another, the second thing pushes back. It therefore seems strange that we have particles that are pushed around by a wave function, while the wave function is completely unaffected by the particles. Of course, quantum mechanics inevitably forces us to confront strange things, so perhaps this consideration should not be paramount.

More important, the original formulations of de Broglie and Bohm both rely heavily on the idea that what really exists are "particles." Just as with GRW, this creates a problem when we try to understand the best models of the world that we actually have, which are quantum field theories. People have proposed ways of "Bohmizing" quantum field theory, and there have been some successes—physicists can be extremely clever when they want to be. But the results feel forced rather than natural. It doesn't mean they are necessarily wrong, but it's a strike against Bohmian theories when compared to Many-Worlds, where including fields or quantum gravity is straightforward.

In our discussion of Bohmian mechanics we referred to the positions of the particles, but not to their momenta. This hearkens back to the days of Newton, who thought of particles as having a position at every moment in time, and velocity (and momentum) as derived from that trajectory, by calculating its rate of change. More modern formulations of classical mechanics (well, since 1833) treat position and momentum on an equal footing. Once we go to quantum mechanics, this perspective is reflected in the Heisenberg uncertainty principle, in which position and momentum appear in exactly the same way. Bohmian mechanics undoes this move, treating position as primary, and momentum as something that derived from it. But it turns out that you can't measure it exactly, due to unavoidable effects of the wave function on the particle positions over time. So at the end of the day, the uncertainty principle remains true in Bohmian mechanics as a practical fact of life, but it doesn't have the automatic naturalness of theories in which the wave function is the only real entity.